

DC Machines & Motors

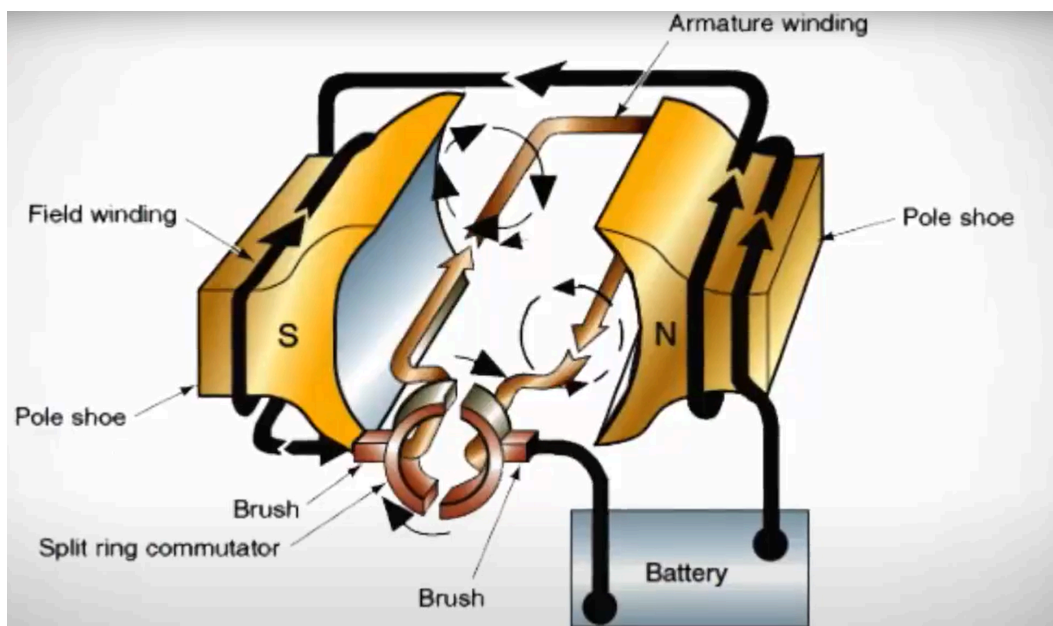
Basic Motor Construction

Generally, a motor consists of stator (stationary housing) and a rotor, the rotating portion that connects to a shaft which couples the motor to the load.

Depending on the type of the motor, there are current carrying coils, called windings, on either the stator, the rotor, or both. These windings are contained in slots, and since the stator/rotor is made of iron, the magnetic fields produced by the windings are intensified. However, the iron used must be laminated to avoid the formation of eddy currents, which can cause energy loss via heat.

Typically, an armature winding produces a field to interact with other stationary fields to produce a torque, thus the current required is dependent on the mechanical load.

Instead of permanent magnets in the standard DC motor, some motors use field windings in place of the permanent magnets to set up magnetic fields, hence the current required is not dependent on the load.



Generally, motors can be classified into AC and DC types. Shown is a list of a few types of motors.

AC motors are motors that are powered from AC sources.

	Type of AC power	Type	Power Range (hp)	Rotor	Stator	Comments and Applications
Ac motors	Three phase	Induction	1–5000	Squirrel cage	Three-phase armature windings	Simple rugged construction; very common; fans, pumps
				Wound field		Adjustable speed using rotor resistance; cranes, hoists
		Synchronous	1–5	Permanent magnet		Precise speed; transport sheet materials
			1000–50,000	Dc field winding		Large constant loads; potential for power-factor correction
	Single phase	Induction	$\frac{1}{3}$ –5	Squirrel cage	Main and auxiliary windings	Several types: split phase, capacitor start, capacitor run; simple and rugged; many household applications: fans, water pumps, refrigerators
		Synchronous	$\frac{1}{10}$ or less	Reluctance or hysteresis	Armature winding	Low torque, fixed speed; timing applications

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DC motors are motors that are powered from DC sources. Rectifier or converter is required to convert AC to DC.

			Power Range (hp)	Armature winding	Field winding	
Dc motors	Wound field	Shunt connected	10–200	Armature winding	Field winding	Industrial applications, grinding, machine tools, hoists
		Series connected				High torque at low speed; dangerous if not loaded; drills, automotive starting motors, (universal motor used for single-phase ac has high power/weight ratio)
		Compound connected				Can be designed to tailor torque–speed characteristic; traction motors
	Permanent-magnet field		$\frac{1}{20}$ –10	Armature winding	Permanent magnets	Servo applications, machine tools, computer peripherals, automotive fans, window motors

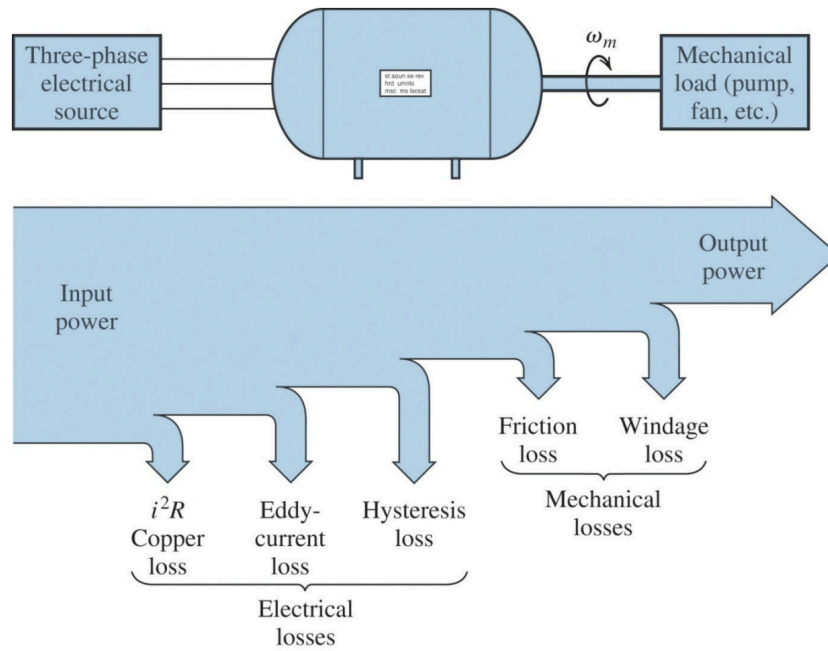
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Note that hp is the symbol for horsepower, and that

$$1hp = 746W$$

Power Ratings & Efficiency

Generally, there are multiple sources that cause an electric motor to be non ideal.



At the output, we can find the output power as

$$P_{out} = T_{out}\omega_m$$

where the angular speed is in rad/s. In rpm, the formula is

$$n_m = \omega_m \times \frac{60}{2\pi}$$

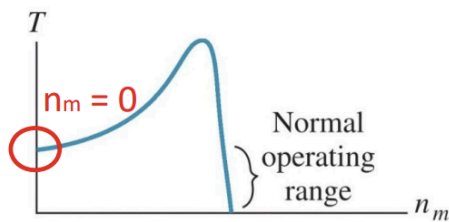
The power rating of a motor refers to the output power the motor can safely produce on a continuous basis. For example, to do work on a mechanical load of 5 hp, the motor must at minimum supply 5hp to the load (hence minimum rating is 5hp)

Hence, the efficiency of a motor is found as

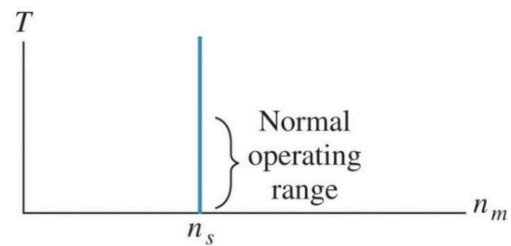
$$\eta = \frac{P_{out}}{P_{in}} \times 100\%$$

Typically, we want to operate motors at 85 to 95% efficiency, as too low will not be sufficient to do work, while too high may lead to overpowering the motor.

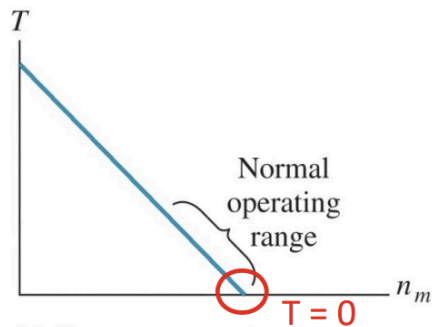
We can also see the different characteristics of motors under different rpms and torques with the torque-speed characteristic graphs:



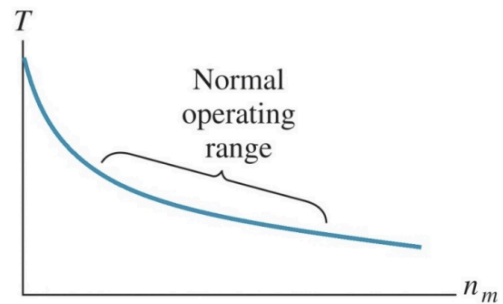
(a) Ac induction motor



(b) Synchronous motor



(c) Shunt-connected or permanent-magnet dc motor



(d) Series-connected dc motor or universal motor

This is where we can better understand the properties of these motors:

AC Induction motor:

Synchronous motor:

As its purpose is to provide a consistent speed, it can only produce a range of torques at a specific rpm, where more power supplied moves up the characteristic graph.

At any other speed, the curve is 0, hence when the motor is not turning, there is no starting torque

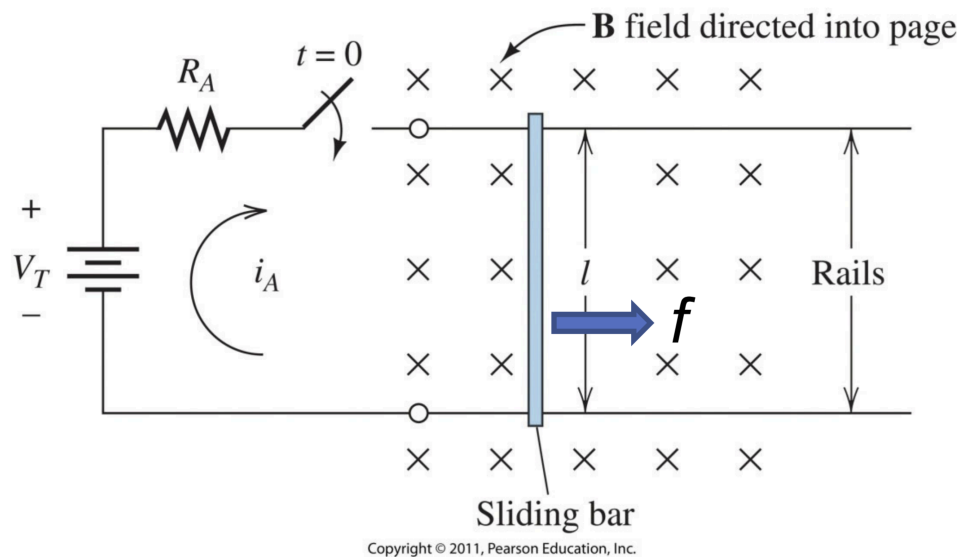
Shunt-connected/permanent magnet DC motor:

Series connected/universal motor

Overall, we pick the correct motor/actuator based on their needed precision/repeatability, the accuracy/resolution of the actuator, the power that needs to be supplied by the motor, and the cost.

Linear DC Machines

Consider the ideal linear DC machine:



The magnetic field and current causes a force that pulls the bar rightward. Since this is a current carrying wire, and B and I are constant and perpendicular, the magnitude of the force is

$$d\mathbf{f} = i d\mathbf{l} \times \mathbf{B} \Rightarrow f = ilB \sin(90^\circ)$$

$$f = ilB$$

As it moves across the rails, it cuts the magnetic flux B , thus a changing magnetic flux is experienced by the bar, hence by $e = \frac{d\lambda}{dt}$, and that $\lambda = N\Phi = NBA$:

$$e = \frac{d}{dt}(NBA)$$

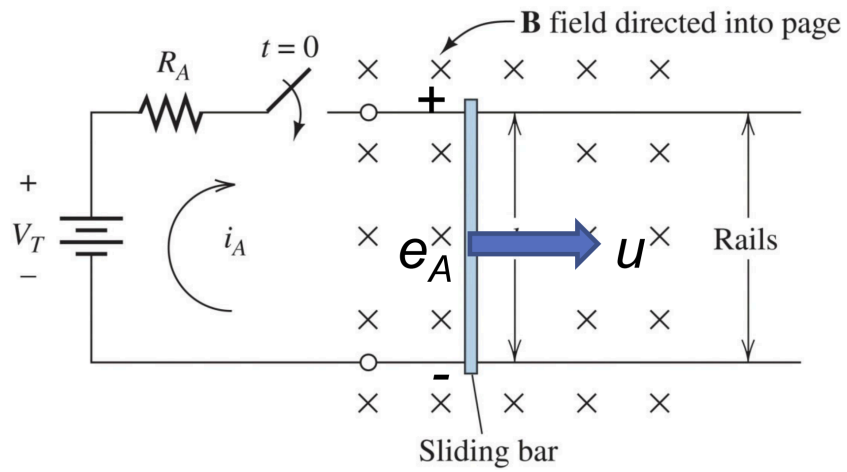
$$= B \frac{dA}{dt} [N = 1]$$

$$= B \frac{d}{dt}(lx)$$

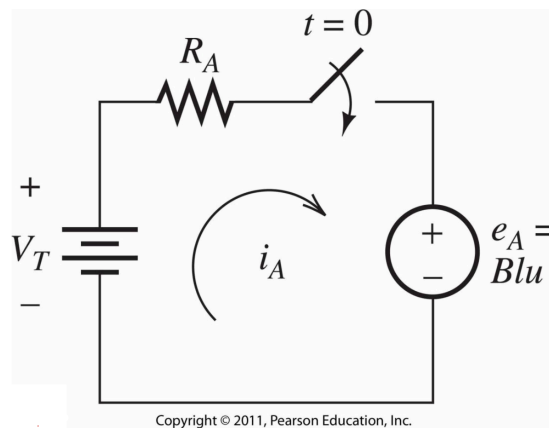
$$= Bl \frac{dx}{dt}$$

$$= Blu$$

In this case, the direction of the voltage produced must resist the original current by Lenz's law, and since current flow from top to down, the voltage induced must produce a current that flows from bottom to top, hence the positive end is on top.



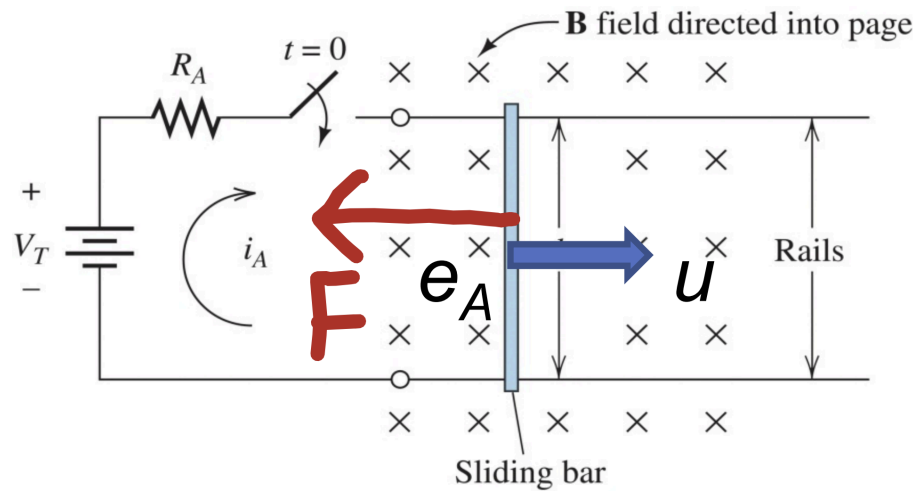
With an expression for the voltage induced in the sliding bar, it can be modelled as a voltage source in series, as



Thus, the current can be found as

$$i = \frac{V_T - e}{R} = \frac{V_T - Blu}{R}$$

Now, say a force F is applied rightward on the slider, like so:



By Newton's Second Law,

$$\begin{aligned}\sum F &= ma \\ f - F &= m \frac{du}{dt} \\ ilB - F &= m \frac{du}{dt} \\ \left(\frac{V_T - e}{R} \right) lB - F &= m \frac{du}{dt}\end{aligned}$$

As we leave the circuit closed, the speed increases and by $e = Blu$ the induced voltage increases too, causing the whole current term to decrease as more voltage is induced to resist V_T , causing f to drop.

At a certain point, $f = F$ and thus $\frac{du}{dt} = 0$, hence u is constant and is in some steady state, say at a value u_{ss} .

Then,

$$\begin{aligned}\left(\frac{V_T - e}{R} \right) lB - F &= 0 \\ \left(\frac{V_T - Blu_{ss}}{R} \right) lB - F &= 0\end{aligned}$$

This steady state speed can be found as

$$\begin{aligned}
\left(\frac{V_T - Blu_{ss}}{R} \right) lB &= F \\
V_T - Blu_{ss} &= \frac{FR}{lB} \\
V_T - \frac{FR}{lB} &= Blu_{ss} \\
\frac{1}{Bl} \left(V_T - \frac{FR}{lB} \right) &= u_{ss} \\
u_{ss} &= \frac{1}{Bl} \left(V_T - \frac{FR}{lB} \right)
\end{aligned}$$

Now, we can see how the system performs under different conditions of F .

No Load Condition

Let $F = 0$ and assume the bar is originally at rest, with the switch turned on at $t = 0$.

Since the bar was originally at rest, we know that $u = 0 \Rightarrow e = 0 \Rightarrow i = \frac{V}{R}$

Hence, this current causes the bar to move rightward, producing a force of $f = ilB$.

This causes the bar to accelerate to u , which causes the voltage $e = Blu$ to be induced and increase as it continues to accelerate.

However, $i = \frac{V-e}{R}$ starts to decrease which causes $f = ilB$ to decrease too.

Soon, the induced voltage reaches $e = V$, which causes $i = 0$ and hence $f = 0$.

This is when the steady state is reached, thus using the values earlier the steady state speed is now

$$\begin{aligned}
u_{ss} &= \frac{1}{Bl} \left(V - \frac{(0)R}{lB} \right) \\
u_{ss} &= \frac{V}{Bl}
\end{aligned}$$

This explains the behavior of motors starting up, where they speed up and accelerate until they reach a certain steady state speed.

Motor Operation

Say a load is applied leftward on the bar, such that it opposes f . In other words, say $F > 0$.

First, we assume that it running at the no-load steady state said earlier, and that F is applied at $t = 0$.

When F is applied, it causes the net force to drop from 0, causing the bar to decelerate, which causes u to decrease. Hence $e = Blu$ drops too.

This causes current $i = \frac{V-e}{R}$ to increase, causing the bar to "react" with an increasing force of $f = ilB$ to balance out the applied load F .

At some point, f increases enough to balance F , reaching the steady state. [$f = F$]

Since some induced force f is present, $i > 0$.

And as for the steady state speed:

$$u_{ss} = \frac{1}{Bl} \left(V_T - \frac{FR}{lB} \right)$$

where it is less than the no load steady state speed of $\frac{V}{Bl}$.

Generator Operation

Say instead a force was applied in the same direction of u , that is $F < 0$, from the same initial conditions as the motor operation

The net force is now positive, hence u increases, causing $e = Blu$ to increase to be greater than V .

Hence, the current $i = \frac{V-e}{R}$ is now negative and flows in the opposite direction and into the voltage source.

As current is now negative, the force $f = ilB$ is now negative and points leftward now.

At a certain point, f increases enough in the opposite direction to balance the applied force F to reach a steady state. Since F is negative, the steady state speed is now

$$u_{ss} = \frac{1}{Bl} \left(V_T + \frac{FR}{lB} \right)$$

where F is the magnitude. This also means that this steady state speed is actually more than the no-load steady speed of $\frac{V}{Bl}$.

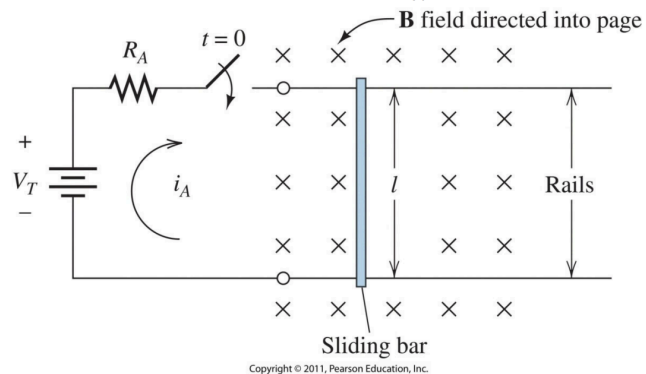
Now, consider the problem:

For the linear machine below, we have $B=1\text{ T}$, $l=0.3\text{ m}$, $V_T=2\text{ V}$, and $R_A=0.05\text{ ohms}$.

(a) Assuming that the bar is stationary at $t=0$, compute the initial current and the initial force on the bar. Also, determine the final (i.e. steady-state) speed assuming that no mechanical load is applied to the bar.

(b) Suppose that a mechanical load of 4 N directed to the left is applied to the bar. In steady state, determine the speed, the power delivered by V_T , the power delivered to the mechanical load, the power lost to heat in the resistance R_A , and the efficiency.

(c) Suppose that a mechanical pulling force of 2 N directed to the right is applied to the moving bar. In steady state, determine the speed, the power taken from the mechanical source, the power delivered to the battery, the power lost to heat in the resistance R_A , and the efficiency.



▼ Solution

(a)

At $t = 0$,

$$i|_{t=0} = \frac{V_T}{R_A} = \frac{2}{0.05} = 40\text{ A}$$

$$\begin{aligned} f|_{t=0} &= ilB \\ &= (40)(0.3)(1) \\ &= 12\text{ N} \end{aligned}$$

When $F=0$,

$$\begin{aligned} u_{ss} &= \frac{V_T}{lB} \\ &= \frac{2}{(0.3)(1)} \\ &= 6.67\text{ m/s} \end{aligned}$$

(b)

The steady state speed can be found as

$$\begin{aligned}
 u_{ss} &= \frac{1}{Bl} \left(V_T - \frac{FR}{lB} \right) \\
 &= \frac{1}{1(0.3)} \left(2 - \frac{(4)(0.05)}{(0.3)(1)} \right) \\
 &= 4.444 \text{ m/s}
 \end{aligned}$$

And as for the current, it can be found as

$$\begin{aligned}
 i &= \frac{V - e}{R} \\
 &= \frac{V - Bl u}{R} \\
 &= \frac{2 - 1(0.3)(4.444)}{0.05} \\
 &= 13.336 \text{ A}
 \end{aligned}$$

Then, the power supplied by the voltage source is

$$P_{in} = V_T i = 2(13.336) = 26.67 \text{ W}$$

As for the output mechanical power, recall that $P = Fv$. Hence,

$$\begin{aligned}
 P_{out} &= fu \\
 &= ilBu \\
 &= (13.336)(0.3)(1)(4.444) \\
 &= 17.77 \text{ W}
 \end{aligned}$$

By conservation of energy, the power dissipated through the resistor is

$$\begin{aligned}
 P_R &= P_{in} - P_{out} \\
 &= 26.67 - 17.77 \\
 &= 8.9 \text{ W}
 \end{aligned}$$

Hence, the efficiency is

$$\begin{aligned}
 \eta &= \frac{P_{out}}{P_{in}} \times 100\% \\
 &= \frac{17.77}{26.67} \times 100\% = 66.63\%
 \end{aligned}$$

(c)

Firstly, note that this case has energy flow from the bar to the battery, as it operating in the generator condition.

The steady state speed can be determined as

$$\begin{aligned} u_{ss} &= \frac{1}{Bl} \left(V_T + \frac{FR}{lB} \right) \\ &= \frac{1}{1(0.3)} \left(2 + \frac{(2)(0.05)}{(0.3)(1)} \right) \\ &= 7.778 \text{ m/s} \end{aligned}$$

and since a force of 2N supplies the energy,

$$P_{in} = fu_{ss} = 2 \times 7.778 = 15.56 \text{ W}$$

Then to find the power supplied to the battery, first find the current as

$$\begin{aligned} i &= \frac{V_T - Blu}{R} \\ &= \frac{2 - (1)(0.3)(7.778)}{0.05} = -6.668 \text{ A} \end{aligned}$$

which means it flows in the reverse direction.

Hence,

$$P_{out} = V_T i = 2(6.668) = 13.336 \text{ W}$$

The power dissipated is then

$$P_R = i^2 R = (6.668)^2 (0.05) = 2.223 \text{ W}$$

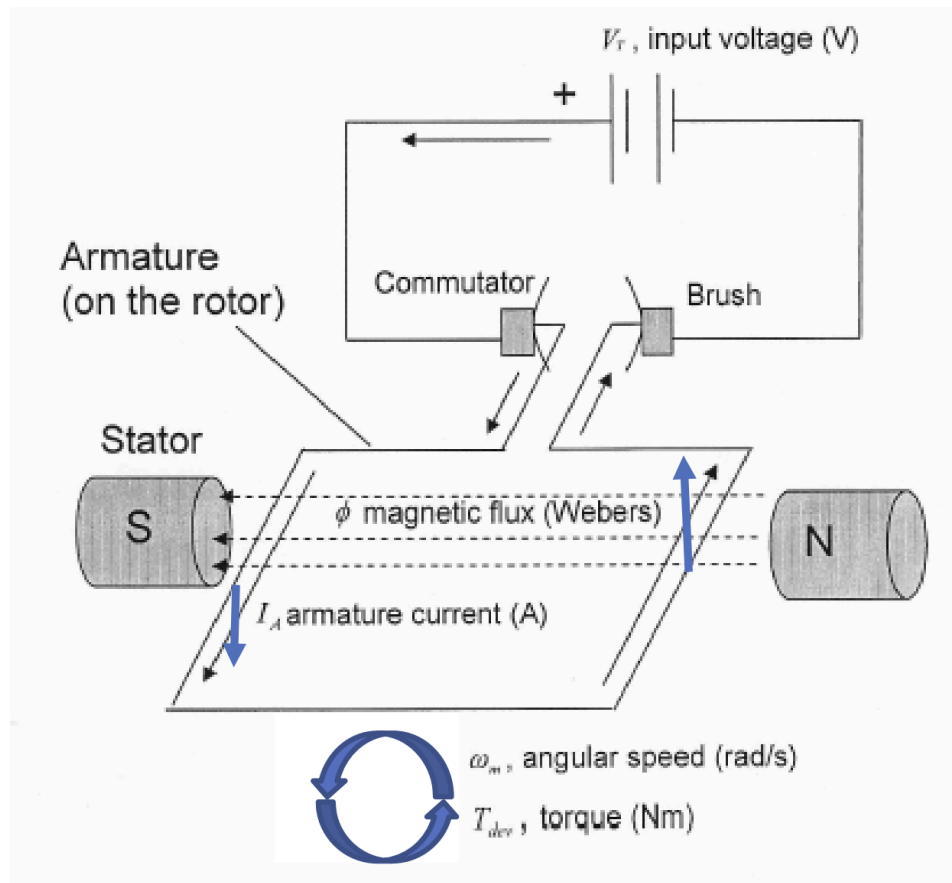
and hence efficiency is

$$\eta = \frac{13.33}{15.56} \times 100\% = 85.71\%$$

Rotary DC Motors

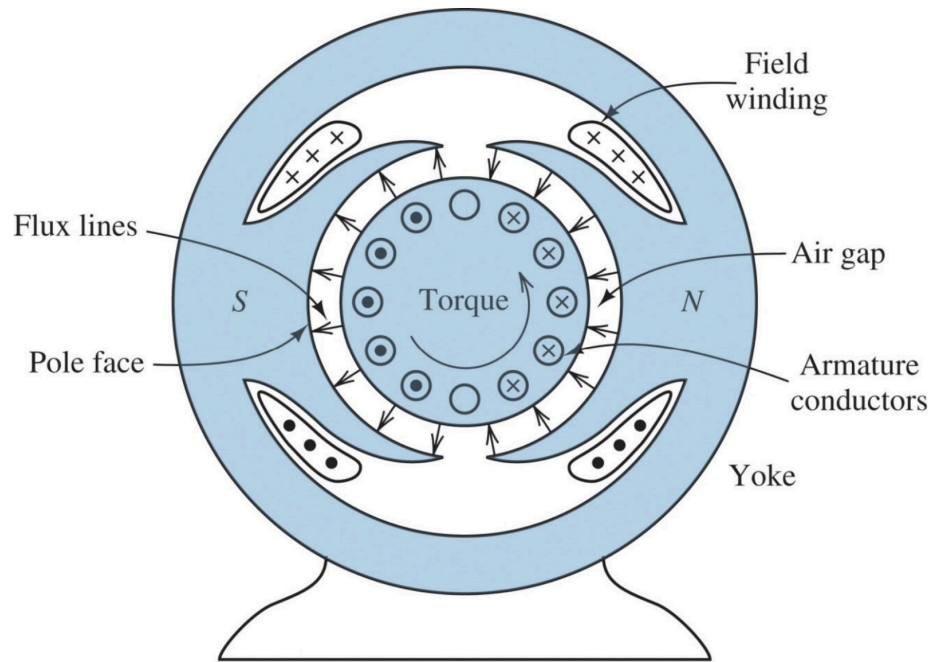
Recall the stator and rotor of a DC motor said in the first section.

The stator provides the magnetic field B like in the linear DC machine, while the rotor is analogous to the sliding bar. i kill myslef



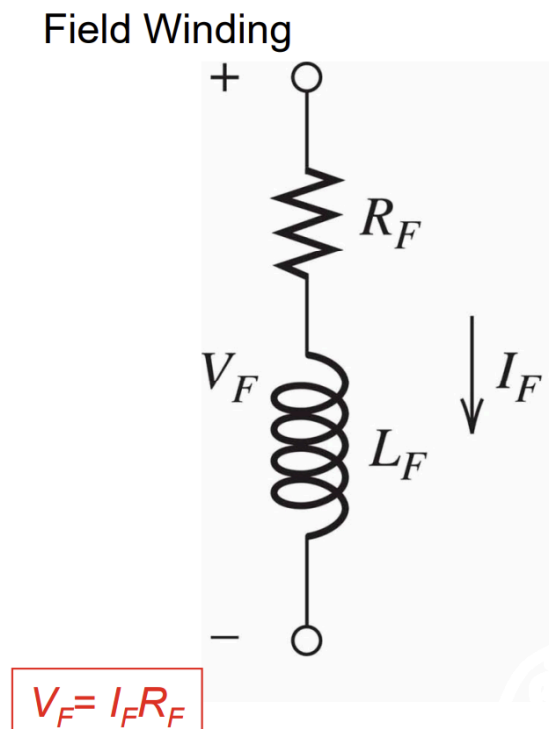
The typical DC motor also requires commutators and brushes, which allow current in the armature to swap to ensure a constant induced voltage in the armature, and hence a continuous rotation of the armature. Typically, more commutators and brushes allow for smoother motion, however require more frequent maintenance.

A double pole switch is used to swap the polarity of the voltage source to control the direction of the motor.



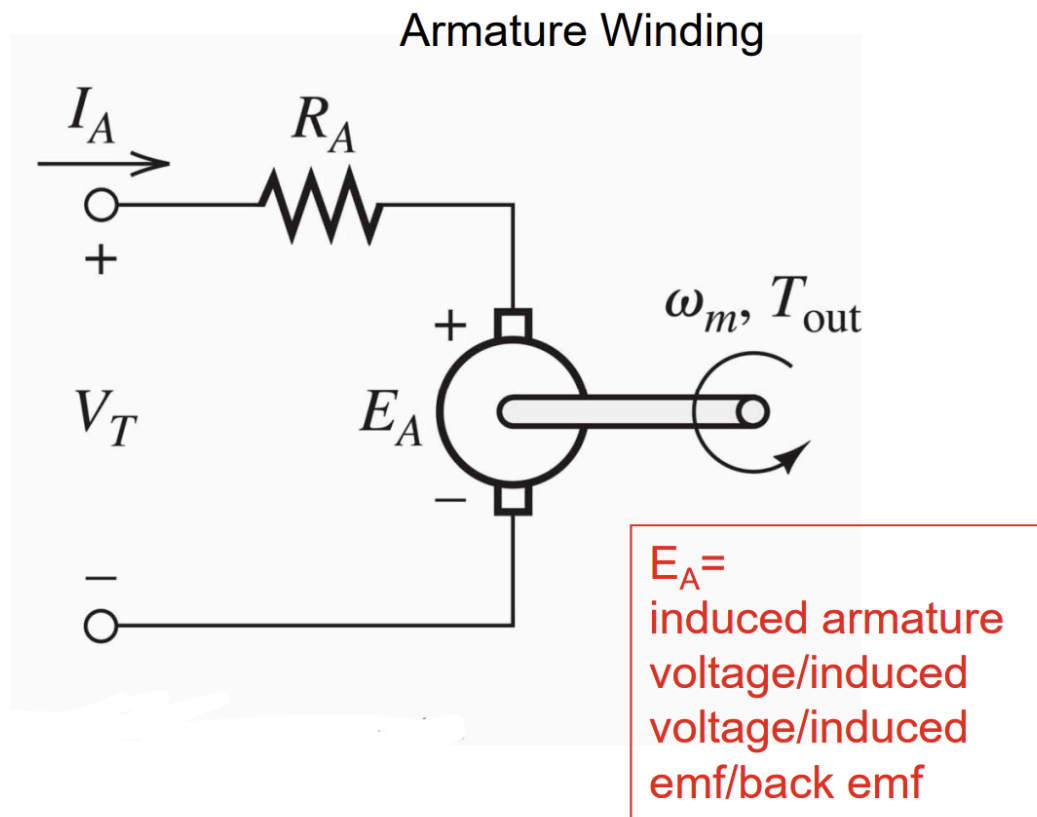
The main working principle is that for each wire in the coil, the magnetic field by the field winding/permanent magnets induce a force that contributes to a turning effect of the armature.

In DC motor analysis, we only care about the steady state of the motors, hence we can represent the field windings with the equivalent circuit:



where the field windings are represented by a solenoid, which will have some inductance. (typically ignored in steady state analysis)

For the armature, it is represented similarly to the linear DC machine:



Notice how the motor does induce some voltage E_A which resists the current. This induced voltage is also called the back emf, the average voltage induced.

If we are given the magnetic flux produced by the stator pole Φ , and some machine constant (to be determined) as K , the back emf can be found as

$$E_A = K\Phi\omega_m$$

Since we can find the power developed in the motor as

$$P_{dev} = E_A I_A = T_{dev} \omega_m$$

where T_{dev} is the developed torque, which is the torque produced by the motor that not only drives a load, but also overcomes rotational losses. This

also means that if rotational losses were 0/negligible, the developed torque is the same as the output torque.

Combining the two equations, we get

$$\begin{aligned} K\Phi\omega_m I_A &= T_{dev}\omega_m \\ K\Phi I_A &= T_{dev} \\ T_{dev} &= K\Phi I_A \end{aligned}$$

For the field winding, the steady state causes the solenoid to act as a wire, hence inductance can be ignored and the current through the solenoid is just

$$I_F = \frac{V_F}{R_F}$$

Ideally, the magnetic flux produced by the field winding is directly proportional to the current, that is

$$\Phi = K_F I_F$$

where K_F is dependent on the physical properties of the winding.

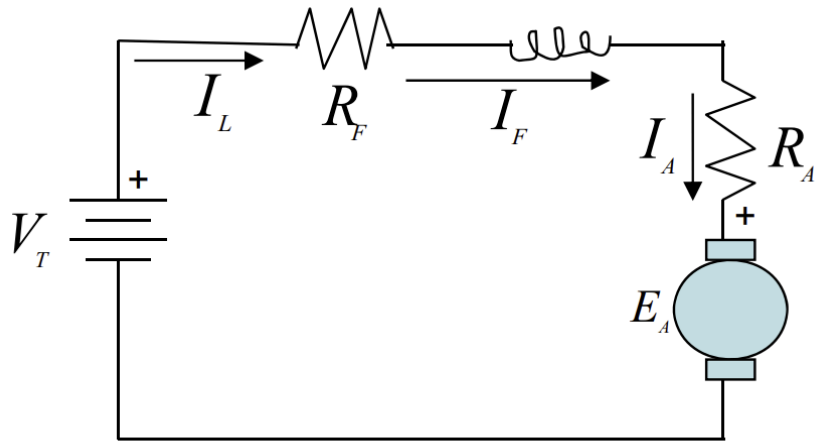
And since E_A is directly proportional to Φ ,

$$\begin{aligned} E_A &= K\Phi\omega_m = KK_F I_F \omega_m \\ \Rightarrow E_A &\propto I_F \end{aligned}$$

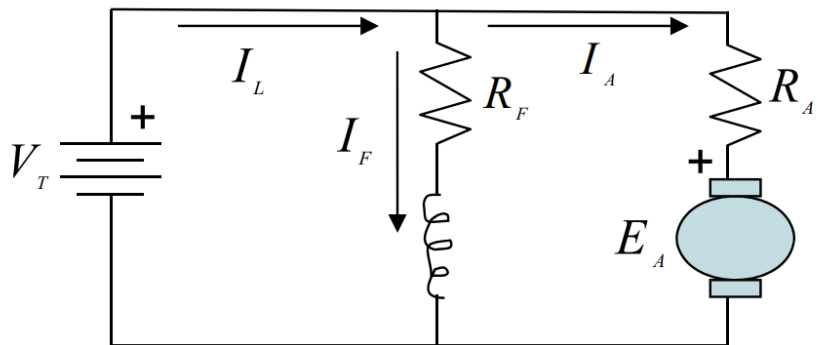
However, due to the hysteresis of electromagnets, as Φ saturates when I_F is very large, E_A will saturate too under the same conditions.

As the voltage source powers both of these windings, we can arrange them in two different ways:

In series, known as DC series motors:



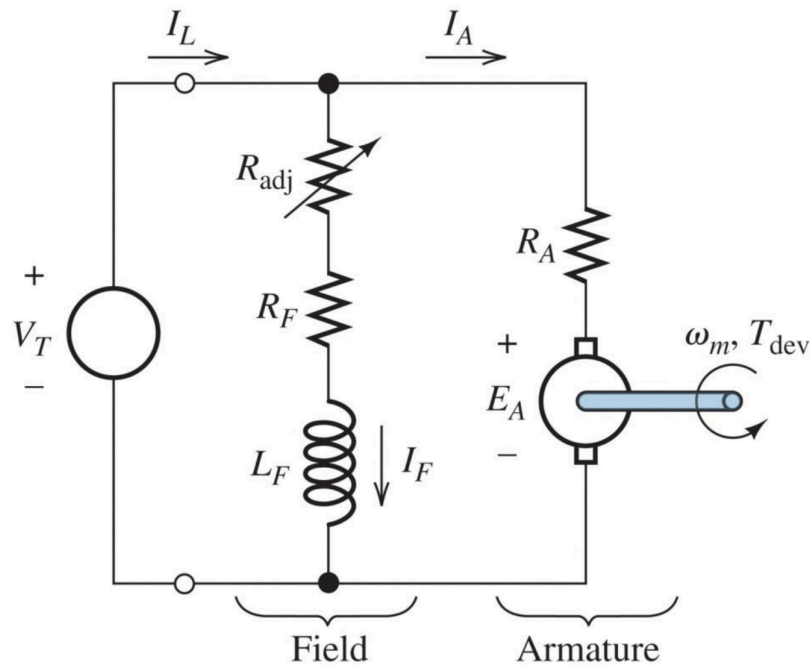
And in parallel, known as DC shunt motors:



A combination of series and parallel windings may be used, to make a DC compound motor.

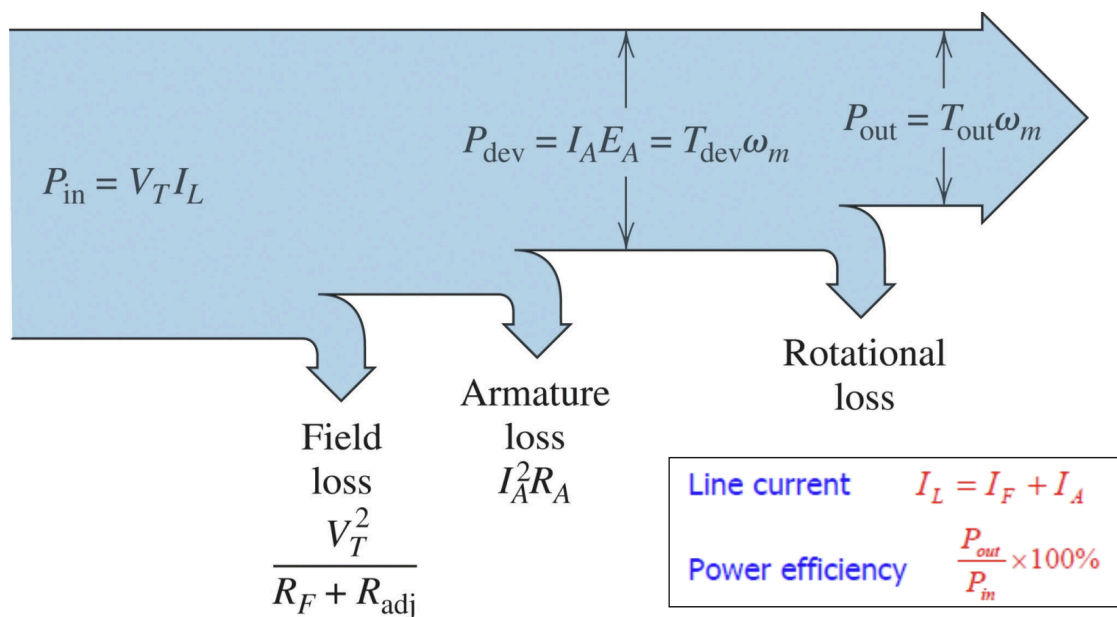
DC Shunt Motors

The full DC Shunt Motor circuit is as shown:



A potentiometer is connected in series with the field winding to adjust the field current and the magnetic flux, which adjusts the speed of the motor.

In a shunt motor, power losses come from the electrical power dissipated from the field and armature windings, and the mechanical rotational loss.



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In other words,

$$P_{in} = P_F + P_A + P_R + P_{out}$$

$$V_T I_L = \frac{V_T^2}{R_F + R_{adj}} + I_A^2 R_A + P_R + T_{out} \omega_m$$

From the circuit diagram, we can find the torque as a function of the angular speed to find the torque-speed characteristic graph.

Applying KVL on the outer loop, we get

$$V_T = I_A R_A + E_A$$

We know that

$$T_{dev} = K \Phi I_A \Rightarrow I_A = \frac{T_{dev}}{K \Phi}$$

and that

$$E_A = K \Phi \omega_m$$

Thus,

$$V_T = \frac{T_{dev}}{K \Phi} R_A + K \Phi \omega_m$$

As we want a function of the developed torque,

$$V_T - K \Phi \omega_m = \frac{T_{dev} R_A}{K \Phi}$$

$$K \Phi (V_T - K \Phi \omega_m) = T_{dev} R_A$$

$$T_{dev} = \frac{K \Phi}{R_A} (V_T - K \Phi \omega_m)$$

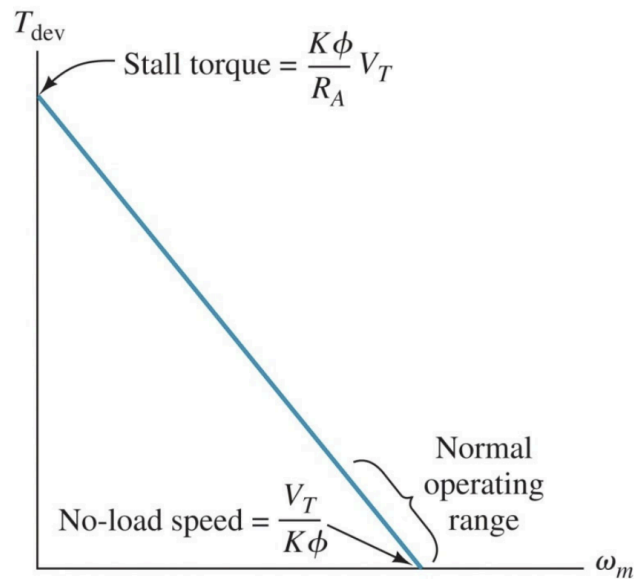
$$T_{dev} = \frac{K \Phi V_T}{R_A} - \frac{K^2 \Phi^2}{R_A} \omega_m$$

In general, the function of developed torque is

$$T_{dev} = a - b \omega_m$$

where a and b are positive constants.

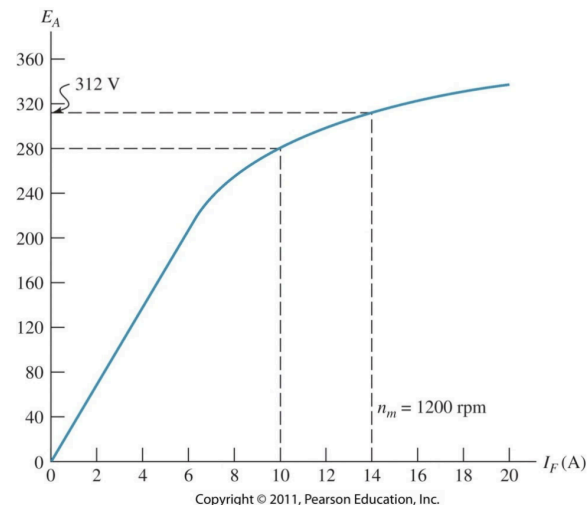
This means that the torque-speed characteristic graph is a straight line with negative slope, starting with an initial torque of $\frac{K \Phi V_T}{R_A}$.



This also means that the x intercept of $\frac{a}{b} = \frac{K\Phi V_T}{R_A} \div \frac{K^2\Phi^2}{R_A} = \frac{V_T}{K\Phi}$ is the speed of the motor under no-load conditions.

Now, consider the problem:

A 50-hp DC shunt motor has the magnetization curve shown below. The DC supply voltage is 240 V, the armature resistance is 0.065 ohms, the field resistance is 10 ohms, and the adjustable resistance is at 14 ohms. At a speed of 1200 rpm, the rotational loss is 1450 W. If this motor drives a hoist that demands a torque of 250 Nm independent of speed, determine the motor speed and efficiency.



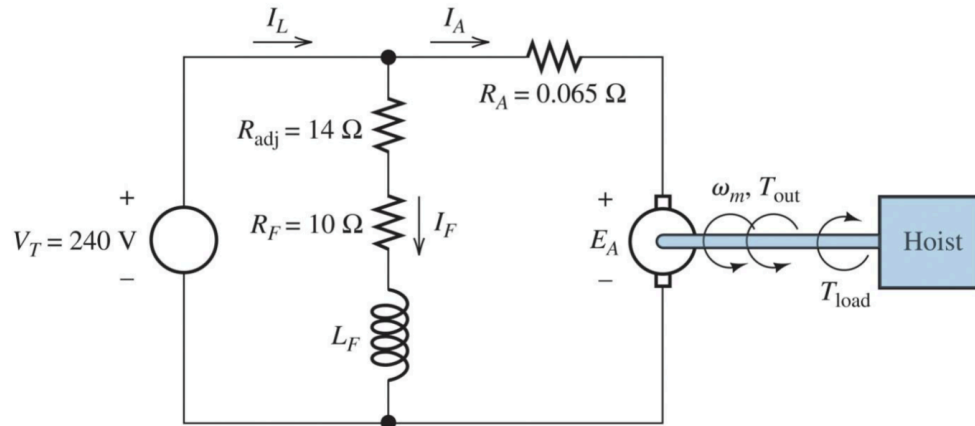
Magnetization curve of the motor

▼ Solution

Firstly, we are given that

$$\begin{aligned}
 P_{out} &= 50hp = 37300W, V_T = 240V, \\
 R_A &= 0.065\Omega, R_F = 10\Omega, R_{adj} = 14\Omega, \\
 T_{out} &= 250Nm \\
 \text{At } n_m &= 1200rpm \Rightarrow P_{rot} = 1450W
 \end{aligned}$$

Equivalent circuit:



Firstly, we can find the field winding's current as

$$\begin{aligned}
 I_F &= \frac{V_T}{R_{adj} + R_F} \\
 &= \frac{240}{10 + 14} \\
 &= 10A
 \end{aligned}$$

Hence, we can determine the induced voltage in the armature from the magnetisation curve, thus

$$I_F = 10A \Rightarrow E_A = 280V$$

We can use this to find $K\Phi$, the "machine constant", as

$$\begin{aligned}
 E_A &= K\Phi\omega_m \\
 K\Phi &= \frac{E_A}{\omega_m} \\
 &= \frac{E_A}{n_m \times \frac{2\pi}{60}} \\
 &= \frac{280}{1200 \times \frac{2\pi}{60}} \\
 &= 2.228
 \end{aligned}$$

We also know the power loss due to the rotation, thus we can find what component of the developed torque is lost due to rotation, as

$$\begin{aligned}
 T_{rot} &= \frac{P_{rot}}{\omega_m} \\
 &= \frac{P_{rot}}{n_m \times \frac{2\pi}{60}} \\
 &= \frac{1450}{1200 \times \frac{2\pi}{60}} \\
 &= 11.54 Nm
 \end{aligned}$$

This means that the motor must develop a torque that overcomes the 11.54Nm loss as well as a load torque of 250Nm, or in other words,

$$\begin{aligned}
 T_{dev} &= T_{rot} + T_{out} \\
 &= 11.54 + 250 \\
 &= 261.54 Nm
 \end{aligned}$$

When we do the previous 2 steps, we are assuming the power loss due to rotation is directly proportional to the speed of the motor, that is $P_{rot} \propto \omega_m$. Hence, this also means that the torque causing the power loss, T_{rot} , is a constant, independent of speed.

Because of this we can use the T_{rot} term from the rotational power loss at 2500rpm, because it will be constant for any speed, and is a constant component of the developed torque.

We can then find the armature current I_A , as

$$\begin{aligned}
 T_{dev} &= K\Phi I_A \\
 I_A &= \frac{T_{dev}}{K\Phi} \\
 &= \frac{261.54}{2.228} \\
 &= 117.38 A
 \end{aligned}$$

Using this, we can find the induced voltage with the outer loop KVL, as

$$\begin{aligned}
 I_A &= \frac{V_T - E_A}{R} \Rightarrow E_A = V_T - I_A R \\
 E_A &= 240 - 117.38(0.065) = 232.37 V
 \end{aligned}$$

Then, the speed at this voltage can be found as

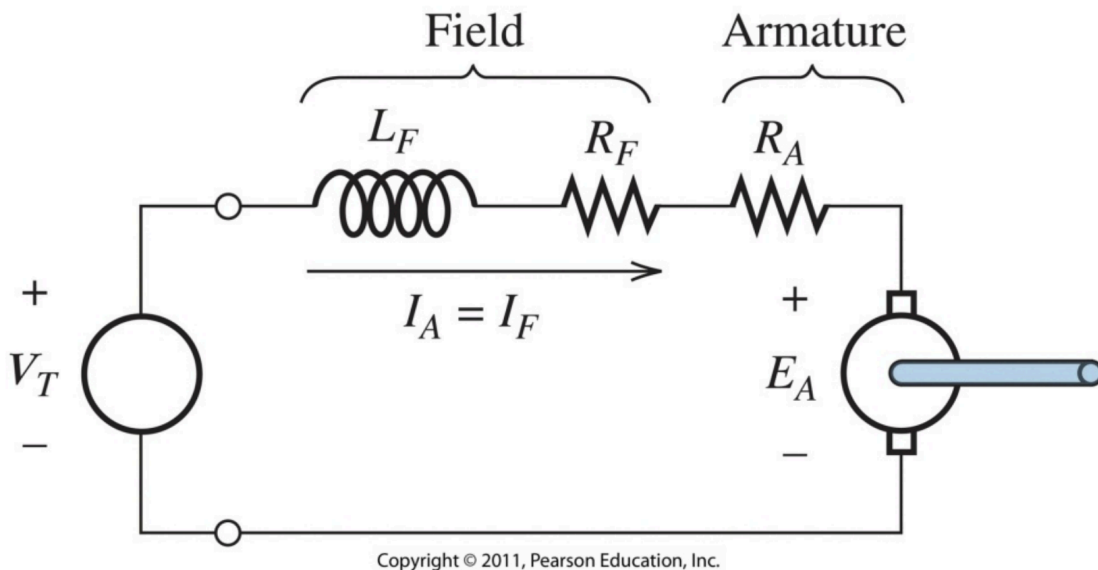
$$\begin{aligned}
 E_A &= K\Phi\omega_m \\
 \omega_m &= \frac{E_A}{K\Phi} \\
 &= \frac{232.37}{2.228} \\
 &= 104.29 \text{ rad/s}
 \end{aligned}$$

From this, the efficiency can be determined by finding the input and output power:

$$\begin{aligned}
 P_{out} &= T_{out}\omega_m = 250(104.29) = 26.072 \text{ kW} \\
 P_{in} &= V_T I_L = V_T(I_F + I_A) = 240(10 + 117.38) = 30.571 \text{ kW} \\
 \therefore \eta &= \frac{26.072}{30.571} \times 100\% = 85.3\%
 \end{aligned}$$

DC Series Motors

As said, series motors have the field and armature winding circuits in series with each other.



As the two are in series, we know that $I_A = I_F$. Hence,

$$\Phi = K_F I_F = K_F I_A$$

From this, we can rewrite the induced voltage and developed torque equations:

$$E_A = K\Phi\omega_m$$

$$\Rightarrow E_A = KK_F I_A \omega_m$$

$$T_{dev} = K\Phi I_A$$

$$\Rightarrow T_{dev} = KK_F I_A^2$$

To find the torque-speed characteristic equation, find use KVL to get

$$V_T = R_F I_A + R_A I_A + E_A$$

$$V_T = R_F I_A + R_A I_A + KK_F I_A \omega_m$$

$$\frac{V_T}{I_A} = R_F + R_A + KK_F \omega_m$$

$$\frac{V_T}{R_F + R_A + KK_F \omega_m} = I_A$$

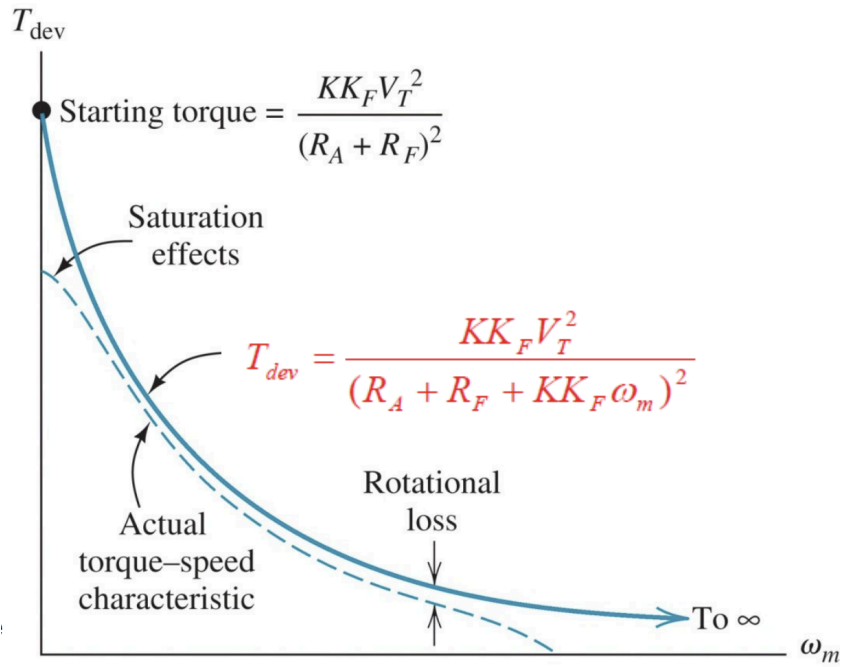
and we substitute this into the developed torque equation we found earlier:

$$T_{dev} = \frac{KK_F V_T^2}{(R_F + R_A + KK_F \omega_m)^2}$$

Notice how this equation is in the form of

$$T_{dev} = \frac{a}{(b + \omega_m)^2}$$

which means that it is a reciprocal graph:



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This means that when the motor is not powered, it has some initial torque of $\frac{KK_F V_T^2}{(R_A + R_F)^2}$, however when $\omega_m = 0$, current is very high and thus the magnetic field produced saturates, making it lesser as it reaches the saturation value instead of the higher ideal value, thus causing a lower torque.

Similarly, under no load conditions ($T_{dev} = 0$), the ideal equation tends to infinite speed, however it is impractical to expect infinite speed from the motor, and thus reaches 0 at a certain point in realistic conditions.

Now, consider the problem:

A DC series motor runs at 1200 rpm while driving a load that demands a torque of 12 Nm. Neglect the resistances, rotational loss, and saturation effects. Find the power output, and then the new speed and output power if the load torque increases to 24 Nm.

▼ Solution

We are given that

$$n_m = 1200 \text{ rpm}, T_{out} = T_{dev} = 12 \text{ Nm}$$

Firstly, the initial power output can be found as

$$P_{out} = \frac{2\pi N}{60} T = \frac{2\pi(1200)}{60} \times 12 = 1508W$$

Then, to find the new speed, we can first use the torque-speed equation and neglect resistances to get:

$$T_{dev} = \frac{KK_F V_T^2}{(KK_F \omega_m)^2}$$

$$T_{dev} = \frac{V_T^2}{KK_F \omega_m^2}$$

$$T_{dev} = \frac{V_T^2}{KK_F} \times \frac{1}{\omega_m^2}$$

If we let $C = \frac{V_T^2}{KK_F}$, we get

$$T_{dev} = C \times \frac{1}{\omega_m^2}$$

Hence, find C when $n_m = 1200rpm$, $T_{dev} = 12Nm$

$$12 = \frac{C}{\left(\frac{2\pi}{60} \times 1200\right)^2}$$

$$C = 12 \left(\frac{2\pi}{60} \times 1200\right)^2 = 189496.4$$

Then, solve for the speed when the developed torque is 24Nm.

$$T_{dev} = C \times \frac{1}{\omega_m^2}$$

$$\omega_m^2 = C \times \frac{1}{T_{dev}}$$

$$\omega_m = \sqrt{\frac{C}{T_{dev}}} = \sqrt{\frac{189496.4}{24}} = 88.85rad/s$$

And hence the new power output is

$$P'_{out} = T_{dev} \omega_m = 24 \times 88.85 = 2132.4W$$